👩🏾‍🏫 Staff Guide · Small Engines Overhaul · Unit 7 Using the Repair Manual (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
His readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as his worksheets for that lesson
- Quick check: score, then scale to his test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Repair manual**: The Briggs & Stratton manual is copyrighted and stays in the classroom; our materials teach how to use it without copying it.
- **iCADEMY Bronze 1-3**: STIHL iCADEMY is login-only; we wrote our own warranty/liability/parts content.

⚠️ Shop safety (with adult supervision)

- **Tools**: students may use all shop tools (hand tools, measuring tools, flywheel puller, valve tools, torque wrench, power tools) **with adult supervision**. Staff model first, stay at the bench, and count tools back in.
- **Eyes and body**: safety glasses on in the shop at all times; coveralls, closed-toe shoes, no loose clothing, long hair tied back, no dangling jewelry.
- **Fuel and oil**: gasoline is flammable. Drain fuel with adult supervision, outside or in a ventilated area, no sparks or flames nearby. Oily rags go in the metal safety can. Gloves for oil; wash hands after.
- **Exhaust**: engines are only started with adult supervision in a well-ventilated area or outdoors (carbon monoxide). Hot mufflers and cylinders burn: let parts cool.
- **Springs and sharp edges**: valve springs and rewind starter springs store energy; use the right tool with adult supervision and eyes protected. Blades, flywheel fins and piston rings can cut.
- **Before working on an engine**: disconnect the spark plug wire so the engine cannot start.
- **Lifting**: engines are heavy; lift with a partner, bend the knees. Car lifts, jacks and 18-wheeler work are video topics in this class.
- **Clean-up**: wipe spills right away (slips), sweep, return every tool to its spot.

📘 Using the Repair Manual: Guided Notes

Small Engines Overhaul · Unit 7 · Level 3 (closer to the class work) · goes with the Manual Reading Handout and parts look-up / warranty topics

Name: _____

Use the class slides and your reading. Key words:

model number *type number* *section* *index* *specification* *exploded diagram* *part number* *warranty* *product liability*

A. Navigation

B. Practice

2. Describe the steps to find the spark plug gap for your engine.

_____

_____

_____

C. Warranty and liability

3. Why do shops follow the manual exactly?

_____

_____

📘 Using the Repair Manual: Reading

Small Engines Overhaul · Unit 7 · Level 3 (closer to the class work) · goes with the Manual Reading Handout and parts look-up / warranty topics

Name: _____

Finding Answers in the Repair Manual

Professional technicians do not guess. They look up procedures and specifications in the manufacturer's repair manual. Mr. Davis's class uses a single-cylinder L-head repair manual. The first step is always to find the engine's model, type and code numbers, usually stamped on the shroud or block, because different versions use different parts and specs.

A manual is organized into sections, one for each system, such as ignition, carburetion and compression. The table of contents lists the sections in order; the index lists topics alphabetically with page numbers. Spec charts are read by finding your model number in the left column and reading across to the value you need, paying attention to units.

To order parts, technicians use exploded diagrams, drawings that show every part spread apart with a reference number, and then read the part number from the parts list. Many manufacturers now offer searchable digital workshop manuals.

Manuals also protect the customer and the shop. A warranty is the manufacturer's promise to repair defects for a set time, but misuse or non-approved parts can void it. Product liability means makers and repair shops can be held responsible if a product causes harm, which is why technicians follow the manual and reinstall all safety guards.

Check your reading

1. Why find the model number first?

2. How do the table of contents and index differ?

3. What can void a warranty?

📘 Repair Manual Look-Up: Lab Sheet

Small Engines Overhaul · Unit 7 · Level 3 · goes with Manual Reading Handout

Name: _____

Question: Can you find specifications and part numbers for your engine using the manual?

Materials

class repair manual, engine model tag, look-up sheet, computer

⚠️ Safety

- Clean hands; no writing in the manual
- Handle the engine with adult supervision

Prediction

Which look-up will be hardest, and why?

_____

_____

Procedure

1. Record model/type/code.
2. Index: spark plug gap.
3. Index: armature air gap.
4. Index: valve clearance.
5. Exploded diagram: air filter part number.

Data

Look-up	Section/page	Spec or part #
Spark plug gap		
Armature air gap		
Valve clearance (intake)		
Air filter part #		

Analysis (Claim, Evidence, Reasoning)

1. Claim: why is the model number the key to every look-up? Evidence: your table. Reasoning.

📘 Using the Repair Manual: Quick Check

Small Engines Overhaul · Unit 7 · Level 3 (closer to the class work) · goes with the Manual Reading Handout and parts look-up / warranty topics

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. The first step before using the manual is…

A. Order parts | B. Start the engine | C. Find the model number

2. Which lists topics alphabetically?

A. Index | B. Table of contents | C. Cover

3. An exploded diagram is used to…

A. Measure compression | B. Mix fuel | C. Find part numbers

4. Using non-approved parts can…

A. Void the warranty | B. Extend the warranty | C. Do nothing

5. Explain product liability in your own words.

6. Why must you check units when reading a spec chart?

📘 Using the Repair Manual: Guided Notes · ANSWER KEY

Small Engines Overhaul · Unit 7 · Level 3 (closer to the class work) · goes with the Manual Reading Handout and parts look-up / warranty topics

Name: _____

Use the class slides and your reading. Key words:

model number | *type number* | *section* | *index* | *specification* | *exploded diagram* | *part number* | *warranty* | *product liability*

A. Navigation

B. Practice

2. Describe the steps to find the spark plug gap for your engine.
Find model number; index 'spark plug'; go to section/page; read the chart row for your model.

C. Warranty and liability

3. Why do shops follow the manual exactly?
To fix it right, keep warranty valid, and avoid liability if something goes wrong.

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

📘 Repair Manual Look-Up: Lab Sheet · ANSWER KEY

Small Engines Overhaul · Unit 7 · Level 3 · goes with Manual Reading Handout

Name: _____

Question: Can you find specifications and part numbers for your engine using the manual?

Materials

class repair manual, engine model tag, look-up sheet, computer

⚠️ Safety

- Clean hands; no writing in the manual
- Handle the engine with adult supervision

Prediction

Which look-up will be hardest, and why?
Often valve clearance (several charts).

Procedure

1. Record model/type/code.
2. Index: spark plug gap.
3. Index: armature air gap.
4. Index: valve clearance.
5. Exploded diagram: air filter part number.

Data

Look-up	Section/page	Spec or part #
Spark plug gap	**Ignition**	**0.030 in**
Armature air gap	**Ignition**	**0.010 in**
Valve clearance (intake)	**Compression/valves**	**0.005-0.007 in**
Air filter part #	**Carburetion**	**varies by model**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: why is the model number the key to every look-up? Evidence: your table. Reasoning.
Each chart is organized by model; specs and parts vary by version.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

📘 Using the Repair Manual: Quick Check · ANSWER KEY

Small Engines Overhaul · Unit 7 · Level 3 (closer to the class work) · goes with the Manual Reading Handout and parts look-up / warranty topics

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. The first step before using the manual is…

A. Order parts | B. Start the engine | **C. Find the model number**

2. Which lists topics alphabetically?

A. Index | B. Table of contents | C. Cover

3. An exploded diagram is used to…

A. Measure compression | B. Mix fuel | **C. Find part numbers**

4. Using non-approved parts can…

A. Void the warranty | B. Extend the warranty | C. Do nothing

5. Explain product liability in your own words.

Makers and shops are responsible if their product or repair causes harm.

6. Why must you check units when reading a spec chart?

Values differ in inches/mm or ft-lb/in-lb; wrong units give wrong settings.

🔑 *Team-written from the Manual Reading Handout; no manual text is copied.*